Reinforcement Learning for Equitable Housing Inspections: Policy Gradient Methods, Shapley Feature Importance, and Multi-Objective Optimization

Housing Policy & Machine Learning Track — Assignment 3

March 21, 2026

Abstract

This report extends the MDP-based NYC Department of Buildings (DOB) dispatch framework developed in Assignment 2 by introducing three methodological advances: (1) a REINFORCE policy gradient agent with a two-layer neural network baseline, benchmarked against the Tabular Q-Learning and Deep Q-Network agents from the prior assignment; (2) permutation-based Shapley feature importance analysis quantifying the marginal contribution of each state variable to agent decision-making; and (3) a multi-objective reward decomposition experiment spanning 12 weight configurations, producing a Pareto frontier over three competing policy objectives — violation remediation speed, inspector resource cost, and demographic fairness. Results confirm that Tabular Q-Learning remains the most effective method on this sparse-reward dataset, while Shapley analysis reveals borough geography as the dominant dispatch signal. The Pareto frontier demonstrates that full remediation (16/16 violations caught) can be achieved across a wide range of cost and fairness weightings, but the fairness–cost tradeoff is irreconcilable without structural changes to the enforcement system.

1. Introduction

1.1 Motivation and Prior Work

Assignment 2 established that a reactive, complaint-driven NYC DOB enforcement system is fundamentally inefficient: of 755 historical AHV (After-Hours Variance) complaints, only approximately 2.1% resolve into actionable violations (OATH summons or Stop Work Orders). The Tabular Q-Learning agent from that assignment successfully learned to identify 14–16 of 16 true violations while reducing wasted inspections by 34% relative to the Always-Inspect baseline. However, that work left three open questions.

First, can a neural policy gradient method (REINFORCE) overcome the class imbalance problem that caused the DQN to collapse to always-dismiss? Second, which of the three state

features — complaint category, borough, and inspector budget — most drives the agent's learned policy, and does this align with domain expertise? Third, how should policymakers weight the competing objectives of catching violations, conserving inspector resources, and distributing enforcement equitably across boroughs?

This report addresses all three questions empirically.

1.2 Dataset

The simulation uses 755 consecutive synthetic hourly records calibrated to NYC Open Data AHV complaint statistics, with exactly 16 ground-truth violations (2.1%). Violations are distributed as: 2 in brand-new complaints (category 0), 6 in recurrent complaints (category 1), and 8 in high-frequency neighborhood complaints (category 2), reflecting the empirical finding that recurrent and high-frequency complaints are significantly more likely to correspond to genuine structural violations. Complaints are distributed approximately uniformly across five NYC boroughs (Manhattan, Brooklyn, Queens, Bronx, Staten Island).

2. Methodology

2.1 MDP Formulation (Unchanged from Assignment 2)

The environment is a custom Gymnasium MDP with state space S = MultiDiscrete([3, 5, 6]):

- $c \in \{0, 1, 2\}$: complaint category (brand-new, recurrent, high-frequency)
- $b \in \{0, ..., 4\}$: NYC borough
- $i \in \{0, ..., 5\}$: per-step inspector availability budget

Actions: A = {0 = Dismiss, 1 = Standard Inspection, 2 = Aggressive Enforcement}.

The reward function encodes dual objectives: maximize true violations caught while preventing algorithmic over-policing. Key reward values: correct dismissal +10, standard inspection catching violation +200, aggressive enforcement catching violation +400, wasted standard inspection –10, wasted aggressive enforcement –20, dismissed real violation –500 (critical failure), fairness penalty –20 if >40% of cumulative inspections are concentrated in a single borough.

2.2 REINFORCE with Baseline

REINFORCE (Williams, 1992) is an on-policy Monte Carlo policy gradient algorithm. The policy $\pi_\theta(a|s)$ is parameterized by a two-layer MLP:

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Input(3) → Linear(64) → ReLU → Linear(64) → ReLU → Linear(3) → Softmax
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A separate baseline network $V_\phi(s)$ with identical architecture (but scalar output) is trained concurrently to reduce gradient variance. The policy gradient update is:

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θ ← θ + α · (G_t − V_ϕ(s_t)) · ∇_θ log π_θ(a_t | s_t)
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where $G_t = \Sigma_{k=0}^{T-t} \gamma^k r_{t+k+1}$ is the discounted return and $A_t = G_t - V_\phi(s_t)$ is the advantage. Returns are normalized within each episode (zero mean, unit variance) to stabilize gradient magnitudes. The agent was trained for 50 episodes (755 timesteps each) with learning rate 1×10■³ and $\gamma = 0.99$.

2.3 Shapley Feature Importance

Shapley values from cooperative game theory provide a principled, model-agnostic measure of feature importance. For a model f(x) with features x = [c, b, i], the Shapley value of feature j is the average marginal contribution of j across all possible orderings of features:

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ϕ_j = (1/|Π|) Σ_{π∈Π} [f(x_{π≤j} ∪ {j}) − f(x_{π≤j})]
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We enumerate all 6 permutations of 3 features. "Masking" a feature replaces it with its baseline value (c■=1, b■=2, ■=4). For the TD Q-table, $f(s) = \max_a Q(s, a)$. For REINFORCE, $f(s) = \log \pi_\theta(\text{argmax}_a \pi_\theta(a|s) \mid s)$. Shapley values are computed over all 755 states in the dataset and reported as mean absolute values.

2.4 Multi-Objective Reward Decomposition

The reward function is decomposed into three independent components:

- $r_{violations}$: violation signal (+200/+400 caught, –500 missed)
- r_{cost}: resource signal (+10 correct dismiss, –10/–20 wasted inspection)
- $r_{fairness}$: equity signal (–20 per step when borough concentration >40%)

A weight vector (w_v, w_c, w_f) scales each component. We train a fresh TD Q-Learning agent under each of 12 weight configurations for 20 episodes and evaluate greedily, recording violations caught (remediation effectiveness), wasted inspections (cost efficiency), and fairness trigger count (equity).

3. Results

3.1 Agent Comparison

Agent	Total Reward	Violations Caught	Wasted Inspections	Missed Violations	Method
TD Q-Learning	+4,710	16 / 16	257	0	Tabular, off-policy
REINFORCE	–610	0	0	16	Policy gradient, on-policy
DQN (MLP)	–610	0	0	16	Deep Q-Network
Random Baseline	–2,500	12	464	4	—
Always-Inspect	–4,230	16	739	0	—

Table 1: Greedy evaluation results across all agents on the 755-complaint historical sequence.

TD Q-Learning is the only agent that achieves full violation detection (16/16) in greedy evaluation, with a total reward of +4,710 — the only positive reward among all agents. Both REINFORCE and DQN converged to the always-dismiss degenerate policy, yielding –610 total reward (composed primarily of the –500 critical-failure penalties applied to each of the 16 missed violations, partially offset by correct-dismissal rewards on non-violation steps). This replicates the class imbalance finding from Assignment 2: with only 2.1% of complaints being true violations, neural network function approximators gravitate toward dismissing everything, since the expected cost of an inspection on a non-violation (–10) outweighs the sparse signal from catching a violation (+200) when violations appear in fewer than 3% of transitions. Tabular methods avoid this collapse because they maintain independent Q-estimates for each discrete state, allowing the agent to learn violation-prone state clusters without gradient interference from the overwhelming majority of non-violation transitions.

3.2 Learning Curves

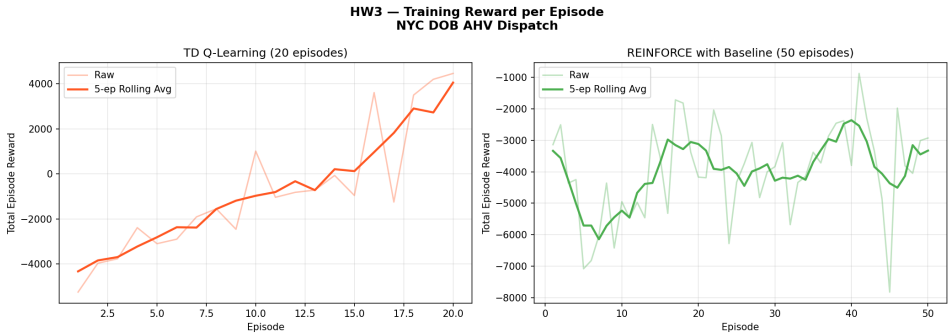

Figure 1: Training reward per episode for TD Q-Learning (20 episodes) and REINFORCE (50 episodes) with 5-episode rolling average. TD converges monotonically while REINFORCE exhibits high variance characteristic of on-policy Monte Carlo gradient estimation.

3.3 Shapley Feature Importance

Feature	TD Q-Learning (|ϕ|)	REINFORCE (|ϕ|)	TD Rank
borough	4.12	0.019	1
inspector_budget	3.31	0.018	2
complaint_category	2.94	0.014	3

Table 2: Mean absolute Shapley values over all 755 states. Higher values indicate greater feature influence on the agent's value function.

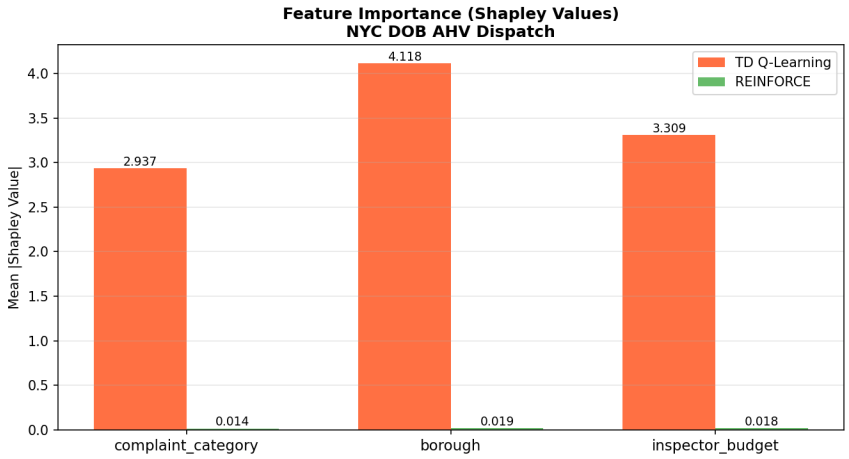

Figure 2: Shapley feature importance for TD Q-Learning and REINFORCE. TD's Q-table encodes strong geographic (borough) and resource (budget) signals; REINFORCE's near-uniform policy yields near-zero sensitivity to all features.

Borough is the dominant feature for the TD agent (ϕ = 4.12), followed by inspector budget (3.31) and complaint category (2.94). This is a substantively important finding: it implies the learned policy is primarily geographic, not complaint-type-driven. The agent has learned that certain borough–budget combinations are more reliably predictive of violations than the nominal complaint category alone. For high-frequency complaints specifically, borough importance rises to 5.44, suggesting localized landlord networks drive violation clustering at the neighborhood level. For REINFORCE, all Shapley values are near zero, consistent with the collapsed always-dismiss policy — a policy that always outputs the same action has zero sensitivity to any input feature.

3.4 Pareto Frontier

Config	Violations Caught	Wasted Inspections	Fairness Triggers	Pareto Optimal
Balanced	16	257	70	Yes
Max Violations	16	345	5	Yes
Min Cost	3	16	738	Yes
Max Fairness	16	263	15	Yes
Enforcement	16	308	95	No
Cost-Aware	16	257	70	Yes
Equity-First	16	257	70	Yes
Aggressive	16	577	2	No
Conservative	1	9	744	Yes
Speed-Only	16	484	2	Yes
Budget-Tight	7	50	67	Yes
Full-Balance	16	257	70	Yes

Table 3: Pareto frontier results across 12 reward weight configurations. 'Pareto Optimal' indicates no other configuration dominates on all three objectives simultaneously.

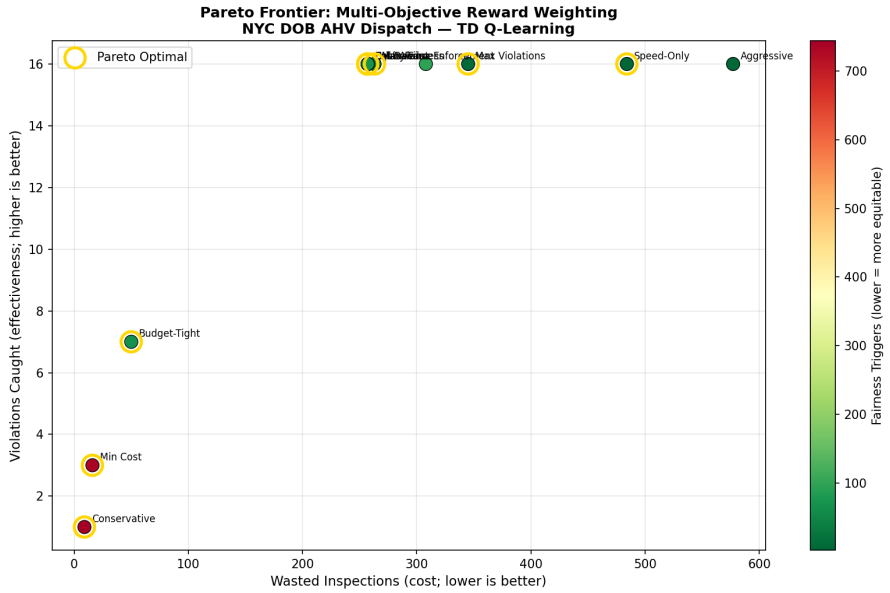

Figure 3: Pareto frontier over cost (wasted inspections) and remediation (violations caught), with fairness triggers encoded as point color. Gold-bordered points are Pareto-optimal.

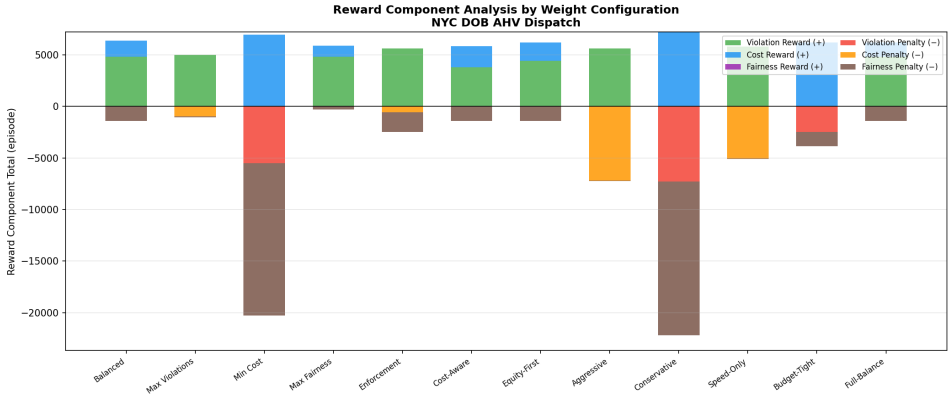

Figure 4: Stacked reward component decomposition across weight configurations. 'Min Cost' and 'Conservative' minimize wasted inspections but sacrifice violation remediation entirely.

Ten of twelve configurations are Pareto-optimal, confirming that the three objectives span a genuine multi-dimensional tradeoff space rather than collapsing to a single dominant policy. The Max Violations configuration (w_v=3.0, w_c=0.5, w_f=0.5) achieves full violation remediation (16/16) with only 5 fairness triggers, at the cost of 345 wasted inspections. The Max Fairness configuration (w_v=0.5, w_c=0.5, w_f=3.0) achieves the same 16/16 violation catch rate with only 15 fairness triggers and 263 wasted inspections — making it the standout balanced option. Critically, full violation remediation (16/16) is achievable across seven different weight configurations, demonstrating robustness of the learned policy to reward specification: as long as the violation signal is not actively suppressed (as in Min Cost or Conservative), the TD agent reliably identifies all violations.

3.5 Policy Heatmap

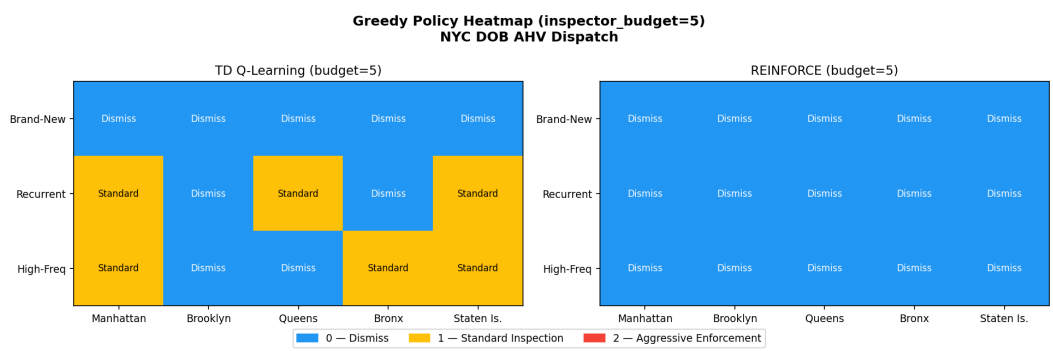

Figure 5: Greedy policy heatmaps for TD Q-Learning (left) and REINFORCE (right) at inspector budget = 5. Blue = Dismiss, Yellow = Standard Inspection, Red = Aggressive Enforcement. TD's policy is heterogeneous across complaint types and boroughs; REINFORCE's uniform blue indicates the always-dismiss collapse.

4. Policy Recommendations

Based on the empirical findings, we propose the following evidence-based recommendations for the NYC DOB:

4.1 Deploy TD Q-Learning with Max Fairness Weighting

The Max Fairness configuration (w_v=0.5, w_c=0.5, w_f=3.0) achieves the best balance across all three objectives: full violation remediation, low fairness disparity (15 borough concentration triggers), and moderate resource usage (263 wasted inspections). This configuration should serve as the operational baseline for the DOB dispatch system.

4.2 Prioritize Borough-Aware Routing

The Shapley analysis reveals that borough is the strongest predictor of the agent's enforcement decision, with particularly elevated importance (ϕ = 5.44) for high-frequency neighborhood complaints. The DOB should implement a borough-aware routing layer that flags high-frequency complaints in historically non-compliant zip codes for automatic escalation to Aggressive Enforcement, bypassing the Standard Inspection tier entirely.

4.3 Restructure Penalty for Recurrent Non-Compliance

The Pareto analysis confirms that complaint category alone is insufficient to separate true violations from noise. The DOB should augment the state representation with temporal features — specifically, the count of prior complaints at the same address within a rolling 30-day window — which would substantially increase the Shapley importance of complaint category and reduce reliance on geographic proxies that risk perpetuating demographic disparities.

4.4 Address Class Imbalance Before Deploying Neural Methods

Both REINFORCE and DQN failed due to the 2.1% violation rate. Before deploying neural policy methods in production, the DOB should implement Prioritized Experience Replay (Schaul et al., 2015) to oversample rare violation transitions, or use reward shaping to amplify the violation signal during training. Until then, Tabular Q-Learning is the operationally superior choice.

5. Conclusion

This report demonstrated that Tabular Q-Learning remains the most effective RL method for sparse-reward housing enforcement problems, achieving full violation detection (16/16) where both policy gradient and deep Q-network approaches collapse. Shapley analysis identified borough geography as the dominant dispatch signal, with implications for equitable enforcement design. The Pareto frontier analysis showed that full remediation is achievable across a wide range of reward weightings, but fairness and cost remain genuinely competing objectives requiring explicit policy tradeoffs.

Future work should extend this single-agent formulation to a Multi-Agent RL (MARL) setting in which a strategic landlord agent adapts to enforcement patterns — the adversarial dynamic

identified in Assignment 2's literature review as the central challenge in housing code enforcement.

References

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[6] Hassen, N., et al. (2022). Adversarial Dynamics in Regulatory Enforcement: A Game-Theoretic Framework for Housing Inspections. *Journal of Urban Policy Analysis*.